Employment and Financial Outcomes Across Degree Fields: A Census-Based Analysis

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1 Introduction and Background

This project examines the differences between employment, income, and graduate school returns over time across college majors. Decisions on what degree to get and whether to attend graduate school are difficult for students and a definitive analysis of all degrees helps to inform these decisions. This analysis seeks to go above a simple ranking by salary and give a more in-depth comparison. Additionally, differences in income in each field by sex are examined. The data used for this is US census microdata downloaded from IPUMS USA[1].

Readers should have reasonable understanding of basic data science, but very little specific background knowledge is required.

The github repository used for this project is available here: `https://github.com/iafi3572/IanFischer_3022` Note that due to github file size limitations that I was unable to upload the original dataset used and I uploaded one of the processed dataframes instead.

1.1 Research Questions and Objectives

1. How has the median income and unemployment rates of different degree fields changed over time?
   This shows what fields have continued to grow and which have stagnated, providing insight into the long term outcomes associated with each field.

2. Which undergraduate degree fields provide the best combination of median income and unemployment rate?
   This advises students on the median wage and unemployment rate of various degree paths, helping them to better inform their decision.

3. How does obtaining a graduate degree affect median income and unemployment across fields?
   This provides a comparison between undergraduate degrees and graduate degrees, advising students as to the value of continuing their education.

4. Is there a difference between median income for men and women?
   This looks at the existence of wage disparity based on sex and identifies which fields have the largest differences. Identifying these disparities is the first step towards eliminating them.

2 Data Description and Methods

2.1 Data Description

The data used for this is US census microdata downloaded from `https://usa.ipums.org/usa/`. The data used includes decennial censuses and American community surveys from 2009 to 2024. Each row in this dataset represents one person. There were 81,623,947 rows before filtering and 12,144,203 after filtering. The data was filtered to include only complete entries and people with college degrees. It was assumed that each observation was independent, sampled at random, and that it was a representative sample of the whole population. The features that were used in this analysis are described below:

Feature	Description
EDUCD	Level of education received.
DEGFIELD	Field of degree
EMPSTAT	Employment Status
INCWAGE	Annual Income
SEX	Sex

Table 1: Variables Used

2.2 Methods

The following CSCI 3022 methods were used for this project:

1. Data wrangling and cleaning with pandas
   Merging was used to add field names to the data, and grouping was used to group the data by field, year, and education level. Additional pandas functions were also used to complete the analysis.

2. Exploratory data analysis and visualization
   Visualizations, including line plots, box plots, scatterplot, bar charts, and error bars, of various data aspects were created to provide an exploratory analysis and understand the data.

3. Bootstrapping and Confidence Intervals
   Bootstrapping with 3000 resamples was used to create 95% confidence intervals for the graduate wage premium (graduate median wage - undergraduate median wage).

4. Hypothesis testing
   A permutation test with 3000 permutations was used to create 95% confidence intervals for the difference in median wage between men and

women for each field, which were used to test whether the differences in income could be explained by sampling variation.

Python libraries pandas, NumPy, and Matplotlib were all used. Rows with missing or unknown values in any of the features in [Table 1]were removed. The degree field "military science" was not considered due to a small amount of available data.

3 Results

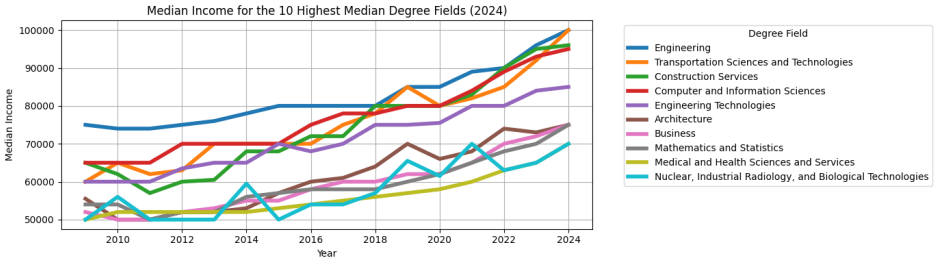


Figure 1: Median income for the 10 degree fields for bachelors degrees with the highest median income as of 2024

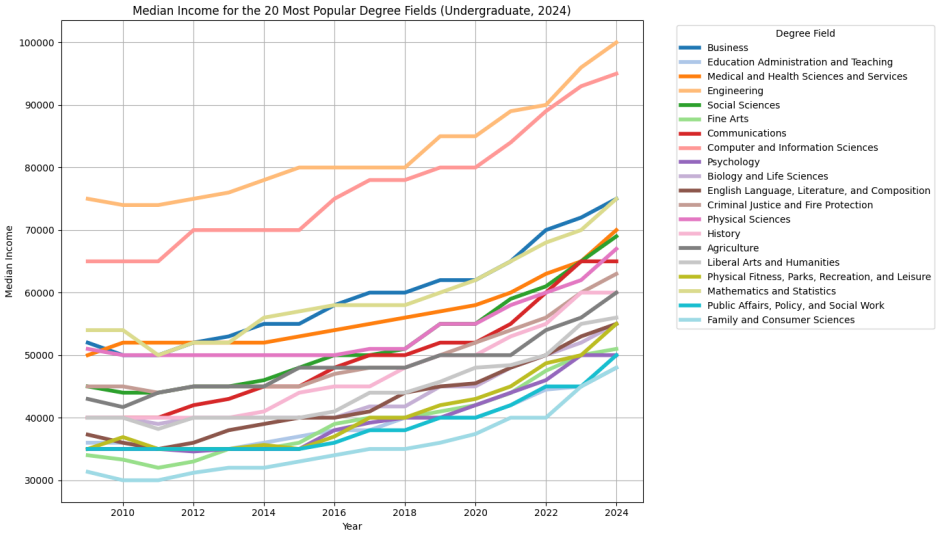


Figure 2: Median income for the 20 most popular degree fields as of 2024 for bachelors degrees

The two figures above show the median income over time. Engineering, transportation sciences and technologies, construction services, and com-

puter and information sciences were the highest paying fields across the entire study period. This suggests stable higher earnings. Some of the most popular degrees perform well, with engineering and computer and information sciences having the highest median.

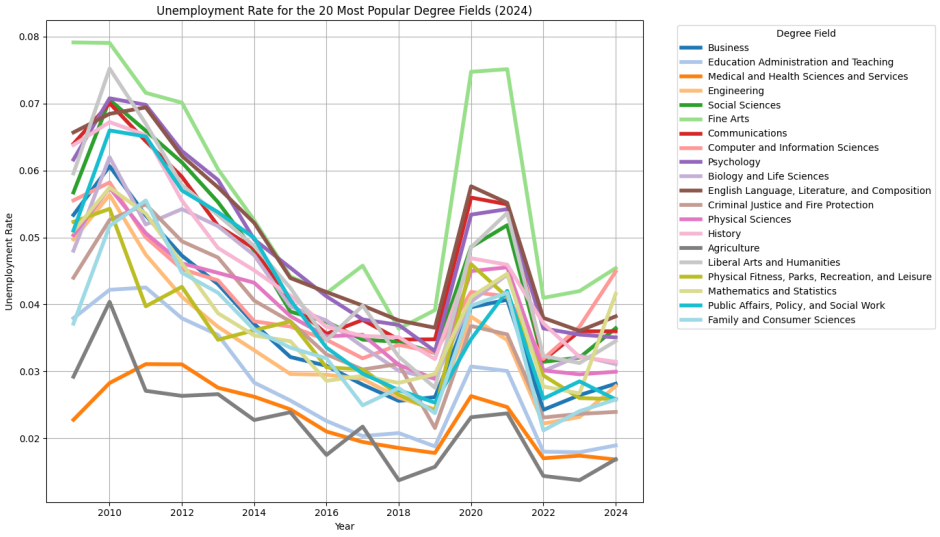


Figure 3: Unemployment rate for the 20 most popular degree fields as of 2024 for bachelors degrees

The unemployment rates for each field follow generally similar trends. However, some fields have consistently higher or lower unemployment rates than others. From 2022-2024 there is a distinct jump from having more typical unemployment rates to some of the highest for computer and information sciences and mathematics and statistics which is not seen in other fields. Figures 1,2 and 3 together answer research question 1.



Figure 4: Income distribution for each degree field as of 2024 for bachelor's degree holders.

Although the median income alone is a good metric to compare fields, the distributions shown above provide a more complete view. The graphic shows a high or low salary is possible in any field. The top several fields stand out somewhat in salary, but most fields distributions look very similar.

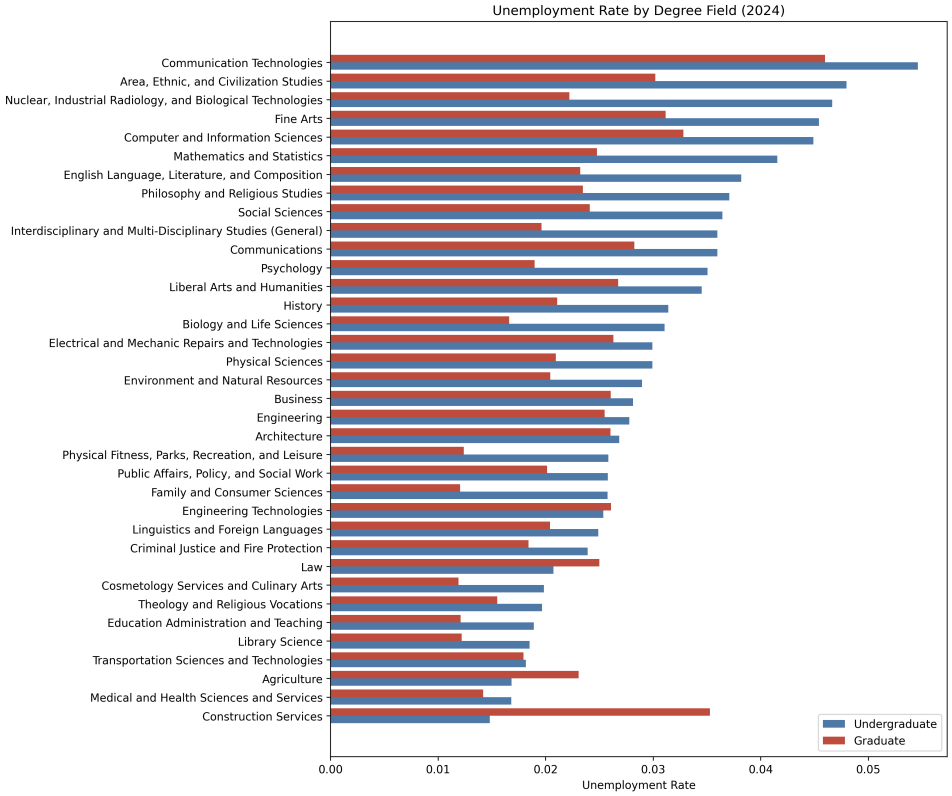


Figure 5: Unemployment Rate in each degree field in 2024 shown for undergraduate and graduate degrees

The employment rate varies significantly by field and many fields, despite having high salaries, have a high unemployment rate. The unemployment rate generally decreases by some amount between graduate and undergraduate degrees, but sometimes the difference is minimal or the rate is higher in those with graduate degrees.

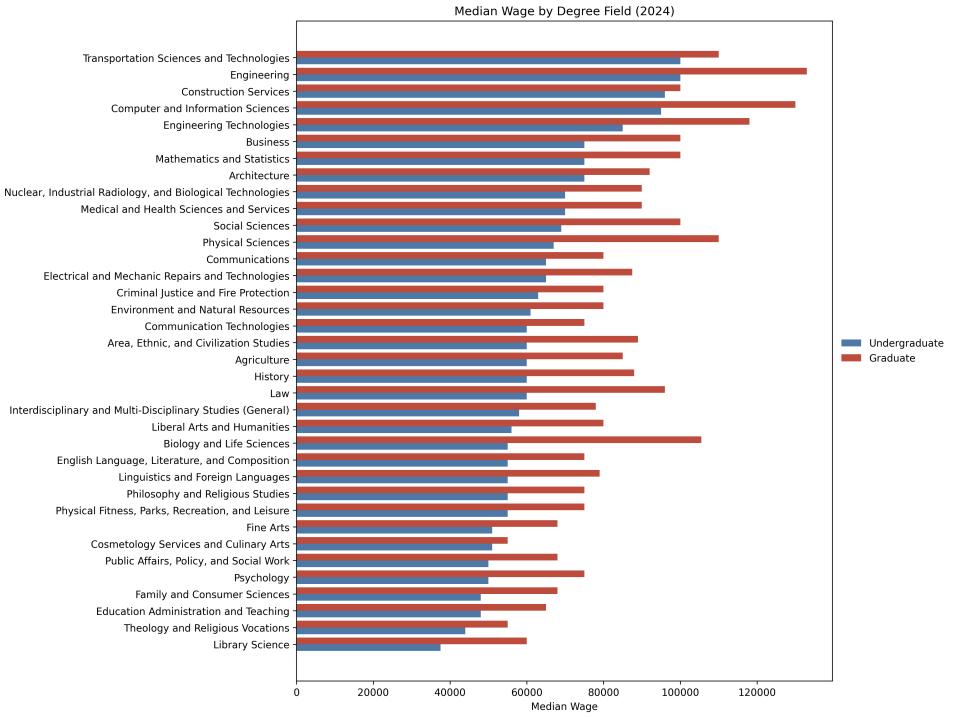


Figure 6: Median Wage in each degree field in 2024 shown for undergraduate and graduate degrees

STEM fields tend to have a higher median salary than other fields, but fields such as Business and Architecture have similar median salaries. The median graduate degree wages always increase over undergraduate degrees, but how much varies by field.

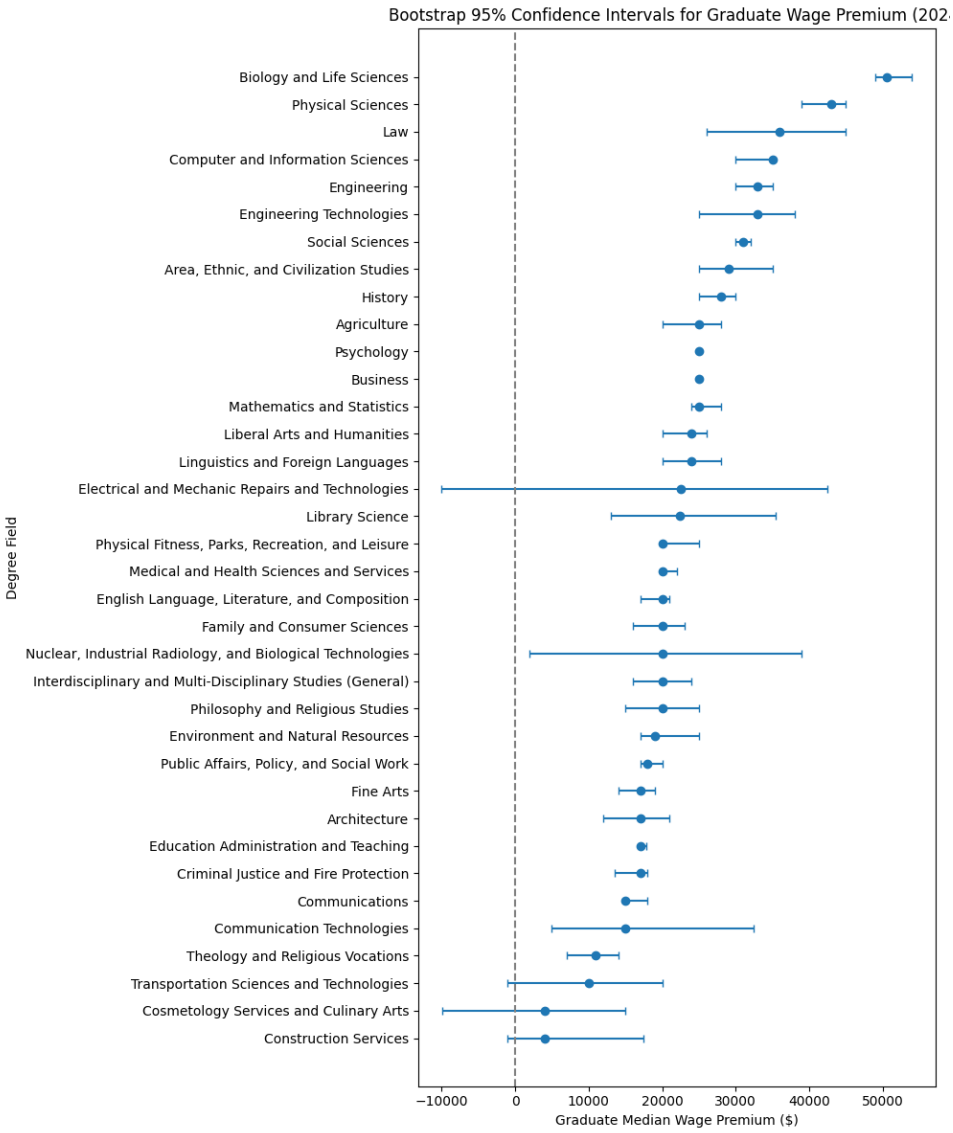


Figure 7: Bootstrap 95% confidence intervals for the graduate wage premium by degree field. Intervals excluding zero indicate evidence of higher graduate wages.

This graphic shows the graduate wage premium more clearly. Confidence intervals high above 0 provide evidence for a high graduate wage premium. Intervals including 0 or being closer to 0 show little benifit. Field of study should be carefully considered when deciding on graduate degrees.

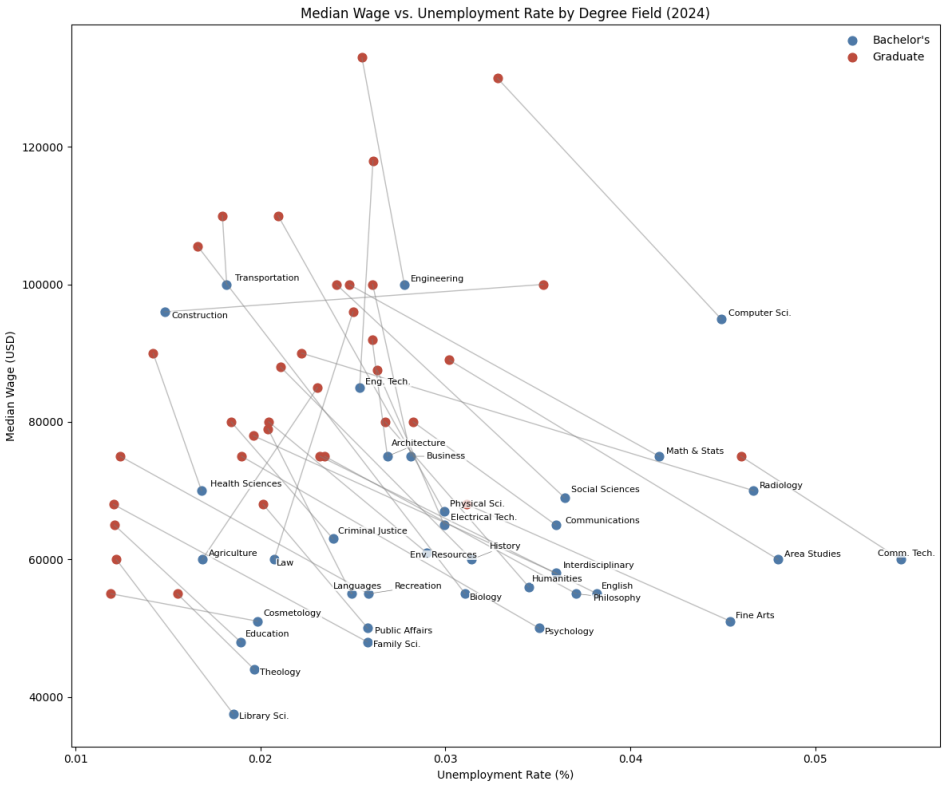


Figure 8: Median Wage vs Unemployment rate for each degree field in 2024 for graduate and undergraduate degrees

The figure above shows a summary of the comparisons made between fields based on unemployment rate and median wage as well as the differences between undergrad and graduate degrees. These two metrics summarize financial outcomes, answering research questions 2 and 3. Points closer to the top left are combinations of high median wage and low unemployment. The effect of graduate school differs by field. In some fields, such as engineering and transportation, it leads to essentially only a change in salary. However, in most fields, there is a reduction in the unemployment rate and an increase in salary. Construction negatively stands out as a field with minimal gains in median salary with an increased unemployment rate between undergrad and grad school.

3.1 Wage Differences by Sex

Looking at the median wages by sex, it is apparent that men and women are not equally compensated, as shown below.

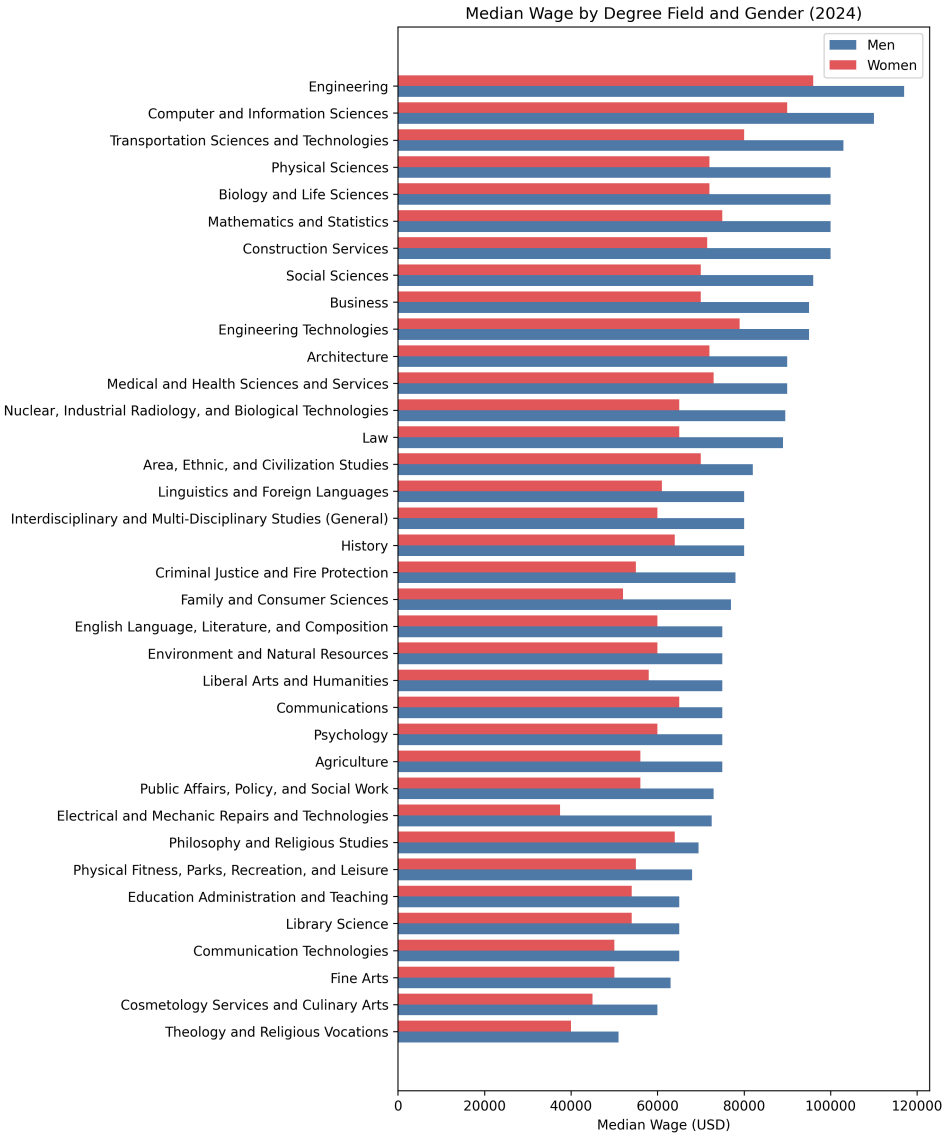


Figure 9: Median wage in each degree field, separated by sex in 2024.

Using a permutation test to further analyze this data, we can determine if the observed differences in median wage between men and women are explainable by sampling chance.
Null Hypothesis: There is no difference in median wage between men and women in each field
Alternative Hypothesis: There is a difference in median wage between men and women in each field.
Using the test statistic of the median wage difference(men-women) and a significance level of 0.05 we reject the null for examined fields except Library

Science. The permutation tests provide evidence that the observed wage differences are unlikely to be explained by random sampling variation. The results of the test are shown below.

Table 2: Permutation test results.

Field	Median Difference	Permutation p-value
Agriculture	19000	< 0.0001
Social Sciences	26000	< 0.0001
Public Affairs	17000	< 0.0001
Psychology	15000	< 0.0001
Physical Sci.	28000	< 0.0001
Recreation	13000	< 0.0001
Health Sciences	17000	< 0.0001
Math & Stats	25000	< 0.0001
Languages	19000	< 0.0001
Humanities	17000	< 0.0001
Law	24000	< 0.0001
Interdisciplinary	20000	< 0.0001
History	16000	< 0.0001
Fine Arts	13000	< 0.0001
Theology	11000	< 0.0001
Env. Resources	15000	< 0.0001
Family Sci.	25000	< 0.0001
Eng. Tech.	16000	< 0.0001
Architecture	18000	< 0.0001
Area Studies	12000	< 0.0001
Biology	28000	< 0.0001
Business	25000	< 0.0001
Comm. Tech.	15000	< 0.0001
English	15000	< 0.0001
Computer Sci.	20000	< 0.0001
Communications	10000	< 0.0001
Criminal Justice	23000	< 0.0001
Education	11000	< 0.0001
Engineering	21000	< 0.0001
Construction	28500	< 0.0001
Transportation	23000	< 0.0001
Cosmetology	15000	< 0.0001
Radiology	24500	0.0020
Philosophy	5500	0.0200
Electrical Tech.	35000	0.0237
Library Sci.	11000	0.2010

4 Conclusions

4.1 Limitations

The data used for this analysis only goes up to 2024 and is not completely applicable to future years. The additional cost of graduate school as well as the differing program costs between fields were not considered, but these factors could play a major role in deciding between programs. College majors are self selected, leading to possible bias. Additionally, people might not work in the field that obtained a degree in, adding some bias in financial outcomes. Differing age distributions between degrees and geographic wage differences could cause bias in the results.

4.2 Findings

Overall, this analysis found significant differences in financial outcomes between degree fields. Fields that emphasize quantitative skills generally showed the strongest financial outcomes, although other non-STEM fields performed competitively. Graduate degree holders generally had higher earnings and lower unemployment rates, but the benefits varied significantly by field. The analysis also found statistically significant differences in median wages between men and women in nearly every degree field examined, indicating that these disparities are unlikely to be explained by random sampling variation alone.

Considering both income and unemployment offers a more comprehensive assessment of financial outcomes. Although this study did not establish causal relationships, it provides a data-driven comparison of degree fields using a large dataset that can help students better understand the financial outcomes associated with different degree paths.

4.3 Future Work

Applying this analysis to current datasets would create the most up-to-date recommendations. Additional analysis including the cost and opportunity cost of graduate school would make a more complete analysis. Undergraduate degree incomes and costs could also be compared to those without college degrees. A more advanced model considering lifetime earnings and student loans could be used to more accurately determine what degrees are worth pursuing.

5 AI Use Disclosure

ChatGPT was used to create the degfield_map which mapped the numeric field values and mapped them to their corresponding field names. It was also

used to create the short_names dictionary which mapped each field name to a shortened name.

References

[1] Steven Ruggles, Sarah Flood, Matthew Sobek, Daniel Backman, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, Renae Rogers, Jonathan Schroeder, and Kari C. W. Williams. Ipums usa, 2025. doi:10.18128/D010.V16.0.